Joonsuk Bae

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Lead analyzer of the first ALICE Run 3 charged-particle jet measurement. Pb/scintillating-fiber calorimetry R&D for the ePIC Barrel Imaging Calorimeter.

Education

Ph.D. Candidate, Physics, Sungkyunkwan University — 2022.03 – expected 2027.02
Advisor: Beomkyu Kim. Thesis: *R-dependent charged-particle jet production in pp at* $\sqrt{s} = 13.6$ *TeV with ALICE.*

B.S., Physics, Sungkyunkwan University — 2021.02

Appointments and Collaboration Membership

ALICE Collaboration, CERN — 2022.10 – present
ePIC Collaboration, Brookhaven National Laboratory — 2024.02 – present
LAMPS Collaboration, RAON — 2022.03 – 2024.04

Research Highlights

- **Lead analyzer**, first ALICE Run 3 charged-particle jet cross-section measurement at $\sqrt{s} = 13.6$ TeV (in preparation, target 2026): end-to-end pipeline from O^2Physics task development through QA, unfolding, and full systematic-uncertainty programme; PWG-JE-approved results establish the Run 3 inclusive charged-jet baseline.
- **PWG-JE tracking-efficiency systematics for Hard Probes 2026**: contributed the pp $\sqrt{s} = 13.6$ TeV charged-particle tracking-efficiency systematic ($\sim$1.5 % across the dominant p_{T} range; p_{T}-binned) across 2022, 2023, and 2024 periods; ITS-TPC matching code adopted by a PWG-JE colleague for PbPb tracking.
- **ALICE O^2Physics framework** (AliceO2Group/O2Physics): 35 pull requests / 19 merged including a central patch in `Common/` (PR #15174, TrackTuner) and sole author of `jetCrossSectionEfficiency.cxx` (PR #14057) used by all Run 3 charged-jet analyses for cross-section normalization.
- **ePIC Barrel Imaging Calorimeter**: lead analyzer for the on-site prompt and follow-up offline analysis of the CERN PS T10 (2025.07) Pb/scintillating-fiber test beam; co-author on the first beam-test paper (submitted, NIMA-D-26-00482) with module-construction support and PMT/module performance characterization.
- **Mentoring of MSc analyzers** (W. Ham + 1) on the ALICE Run 3 O–O charged-jet measurement: jet reconstruction methodology, correction procedures, and systematic-uncertainty estimation; co-author on the corresponding Analysis Note with service-level normalization contribution.

Selected Publications

Full list at inspirehep.net/authors/2117232; ALICE Collaboration co-author since 2022.10 **(109 collaboration papers)**.

- FLAGSHIP ALICE Collaboration, *R-dependent charged-particle jet production in pp collisions at* $\sqrt{s} = 13.6$ *TeV with ALICE* (in preparation, target 2026). *[lead analyzer; first ALICE Run 3 charged-jet publication; thesis paper]*
- Korea-BIC Collaboration, *Beam-test characterization of a Pb/SciFi BIC prototype — CERN PS T10 (2025.07)* (in preparation, 2026). *[lead analyzer; prompt and follow-up offline analysis]*
- H. Lee, C. Lee, J. Ryu, G. An, **J. Bae** *et al.*, "Beam test of a Pb/SciFi prototype for the BIC at the EIC," submitted to *Nucl. Instrum. Meth. A* (NIMA-D-26-00482; arXiv preprint). *[co-author; module support and PMT/module performance characterization]*
- Y. Hong, J. Bok, G. An, **J. Bae** *et al.*, "Test-beam performance of the AstroPix silicon sensor for imaging calorimetry," submitted to *Nucl. Instrum. Meth. A* (NIMA-D-26-00572). *[co-author; ePIC BIC silicon-layer test beam]*
- J. K. Adkins *et al.* (ECCE Coll.), "Design of the ECCE detector for the EIC," *Nucl. Instrum. Meth. A* **1073** (2025) 170240. *[co-author; ECCE consortium]*
- Y. J. Kim *et al.* (LAMPS), "Large Acceptance Multi-Purpose Spectrometer (LAMPS) at the RAON," *J. Korean Phys. Soc.* **87** (2025) 670–682. *[co-author; Time-of-Flight maintenance and on-site checks]*

Invited and Contributed Talks (since 2022)

Summary: 3 invited workshops · 5 invited collaboration-meeting talks · 6 international conference contributions (one on behalf of the ALICE Collaboration at EPS-HEP 2025) · 8 domestic (KPS) · PWG-JE internal: 50+.

Invited (external workshops, seminars).

- **FKPPN Workshop**, Inha University — *Cross-section normalization in Run 3; charged-jet studies in ALICE Run 3* — 2025.12

- **HIM 2025**, APCTP — *From precision to structure: jet measurements in ALICE Run 3 and ongoing substructure studies* — 2025.05
- **FKPPN Workshop**, Inha University — *Charged-jet studies in ALICE Run 3* — 2023.11

Invited collaboration-meeting talks.

- **Korea-BIC 2025 Summer Workshop** — *Data analysis of the recent beam test at CERN* — 2025.09
- **KoALICE National Workshop 2024** — *Jet analysis status: first look at the charged-particle jet spectrum in Run 3 pp collisions* — 2025.01
- **ALICE Physics Week 2024** — *Charged-particle jet production in Run 3* — 2024.12
- **KoALICE National Workshop 2023** — *Charged-jet O^2 MC and data QA* — 2024.01
- **KoALICE National Workshop 2022** — *Dijet studies with ALICE at the LHC* — 2023.01

Contributed talks and posters.

- **EPS-HEP 2025**, Marseille *[on behalf of the ALICE Collaboration; PWG-JE merge talk]* — *Inclusive and semi-inclusive jet cross-section measurements in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* — 2025.07
- **INPC 2025**, Daejeon *[international, talk]* — *Charged-particle jet and jet quenching measurement in pp collisions at $\sqrt{s} = 13.6$ TeV during LHC Run 3 with ALICE* — 2025.05
- **Quark Matter 2025**, Frankfurt *[international, poster]* — *Inclusive charged-particle jet spectrum in pp collisions in Run 3 at $\sqrt{s} = 13.6$ TeV with ALICE* — 2025.04
- **Hard Probes 2024**, Nagasaki *[international, poster]* — *First measurements of charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* — 2024.09
- **Quark Matter 2023**, Houston *[international, poster]* — *Charged-particle multiplicity distribution in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* — 2023.09
- **ATHIC 2023**, Incheon *[international, poster]* — *Dijet studies at the LHC* — 2023.04
- **KPS Fall 2025** *[domestic, talk]* — *Inclusive and semi-inclusive jet cross-section measurements in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* — 2025.10
- **KPS Spring 2025** *[domestic, talk]* — *Inclusive charged-particle jet spectrum in pp collisions in Run 3 at $\sqrt{s} = 13.6$ TeV with ALICE* — 2025.04
- **KPS Spring 2025** *[domestic, poster]* — *Prototype calorimeter QC for the Barrel Imaging Calorimeter at the EIC* — 2025.04
- **KPS Fall 2024** *[domestic, talk]* — *First jet measurement in ALICE Run 3: charged-particle jet production* — 2024.10
- **KPS Spring 2024** *[domestic, talk]* — *Charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV* — 2024.04
- **KPS Fall 2023** *[domestic, talk]* — *First look at charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE Run 3 data* — 2023.10
- **KPS Spring 2023** *[domestic, poster]* — *Dijet studies at the LHC* — 2023.04
- **KPS Fall 2022** *[domestic, poster]* — *Dijet studies with ALICE at the LHC* — 2022.10

Software and Computing

ALICE O^{2}Physics (AliceO2Group/O2Physics, 2023 – present). 35 PRs / 19 merged, including:

- **Central framework patch in** `Common/` (PR #15174, merged) — TrackTuner Q/p_{T} smearing fix.
- **Sole author of** `jetCrossSectionEfficiency.cxx` (PR #14057, merged) — PWG-JE task used by all Run 3 charged-jet analyses for normalization.
- Code review on 5 community PRs across PWG-JE and Common.

Detector and Test-beam Experience

- **ePIC Barrel Imaging Calorimeter** — on-site at **CERN PS T10** (2025.07; analysis lead) and **KEK AF–AR** (2025.03; DAQ shifter, prompt analysis); Pb/scintillating-fiber module-construction support and PMT/SiPM-level QA across all prototype modules; JANA2 / EIC-recon reconstruction studies for sampling-fraction and energy response.
- **ALICE ITS2 alignment monitoring** (CTF level) — service work providing period-by-period tracking-performance diagnostics from compressed time-frame data, below the standard AO2D analyzer interface.
- **LAMPS Forward Tracker** (2022 – 2024) — MPPC/scintillator-plane forward-tracking concept design and Geant4 feasibility simulations.

Service and Mentoring

- **Student mentoring** (non-supervisory) — W. Ham and one further MSc co-analyzer on the ALICE Run 3 O–O charged-jet measurement: jet reconstruction methodology, correction procedures, and systematic-uncertainty estimation; co-author on the corresponding Analysis Note.

- **KoALICE O^2 Tutorial** (2024.01.22–23, SKKU; co-organizer with H. Lee and listed Indico contact) — two-day Run 3 analysis-framework tutorial for the Korean ALICE community; instructor on *Running O^2 Physics with Hyperloop* (solo) and on *PWG-JE practical tasks*.
- **ePIC Korea-BIC analysis coordination** — primary liaison between the SKKU group and the Korea-BIC consortium for beam-test data products.

Technical Skills

Programming. C++, Python, ROOT, Geant4, JANA2 / EIC-recon, PYTHIA, FastJet, Bash, Git, LaTeX.
Analysis. jet reconstruction and unfolding (Bayesian, SVD), efficiency and systematic-uncertainty evaluation, data/MC validation, test-beam calibration and energy-response fitting.
Detector hardware. PMT and SiPM readout characterization, cosmic-ray and source calibration, optical-coupling QA, calorimeter module assembly.
Languages. English (fluent), Korean (native).

References

Available on request.